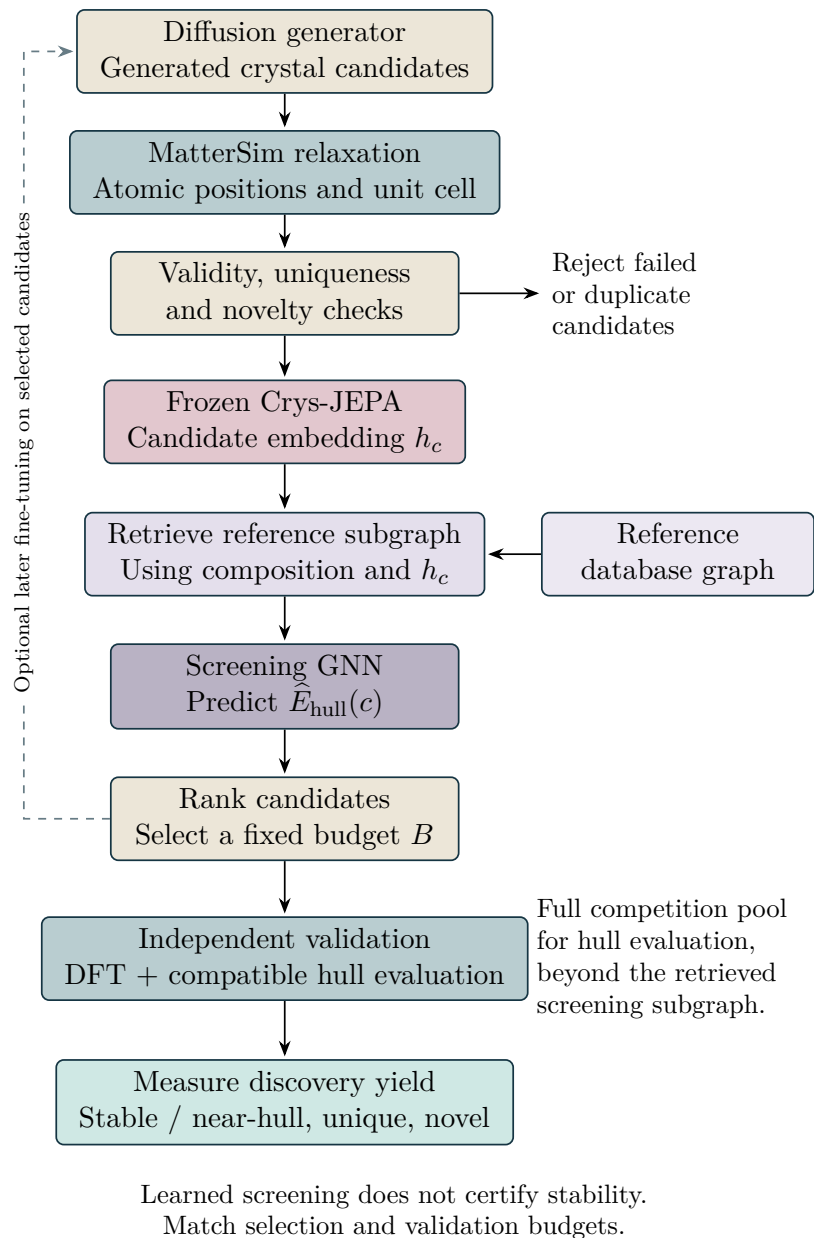
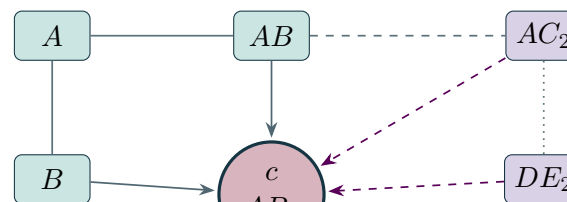


(a) Candidate screening and validation

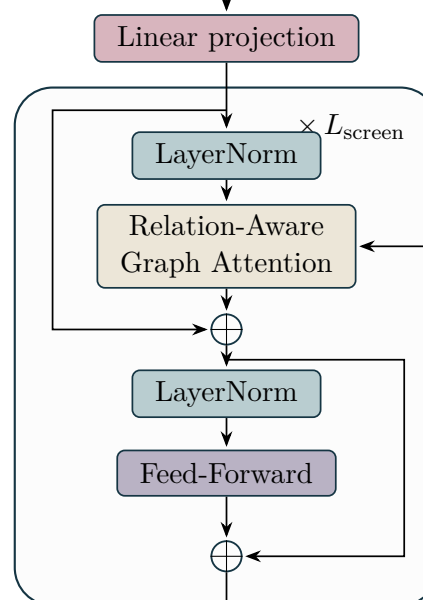


(b) Candidate-conditioned screening GNN

Relaxed candidate c and retrieved whole-crystal references



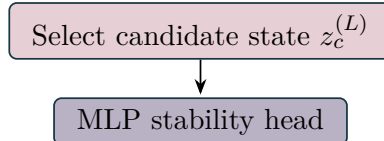
Pink: candidate query node
Teal: competition references
Purple: cross-chemistry analogues
Solid links: chemical context
Dashed links: analogue context
Dotted links: structural similarity



For candidate and reference nodes:
 $x_i = [h_i^{\text{Crys-JEPA}}; \text{composition}_i]$

E, τ_{ij}, e_{ij}
Edges, relation types and similarities

Reference nodes exchange information. The candidate receives competition and analogue context.



Train on labelled crystal queries.
True E_{hull} is a target, not an input feature.
Lower predictions rank first.
Use separate GNN weights for screening and generation.

$$\hat{E}_{\text{hull}}(c) \in \mathbb{R}$$

Proposed flow. Energy-above-hull regression is illustrated; the objective and graph-attention cell remain design choices.